

(ECS708) Report lab3: Regression

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Q5 - What conclusion if any can be drawn from the weight values? How does sex and BMI affect diabetes disease progression? What are the estimated disease progression values for the below examples?

AGE	SEX	BMI	BP	S1	S2	S3	S4	S5	S6
25	F	18	79	130	64.8	61	2	4.1897	68
50	M	28	103	229	162.2	60	4.5	6.107	124

In a linear regression model, each coefficient "w" represents how much the predicted diabetes progression changes. Therefore, the weights represent the influence of each feature on diabetes progression. Positive weights indicate a direct relationship, while negative weights indicate an inverse relationship. In other words, the magnitude of the weights helps identify which features are most important for prediction. For example, features with weights close to zero have a small effect.

The model weight values are: `tensor([[ 1.9413, -11.4535, 26.3020, 16.6372, -10.1897, -2.1343, -7.4969, 8.3693, 22.0618, 2.6037, 153.7365]])`.

BMI has the largest positive weight, meaning that it's a strong predictor of disease progression. The weight being 26.30 is strongly positive, meaning that a higher BMI increases the predicted diabetes progression. Every unit increase in BMI contributes +26 to disease progression.

SEX has a negative weight (-11.45), indicating that higher values in this feature are associated with lower progression. Knowing that female is encoded as 1 and male is encoded as 2, being a male is associated with lower progression compared to being a female.

The estimated disease progression values for the examples are calculated using the linear regression equation:

$$\begin{aligned}\hat{y}_1 &= (1.9413 \times 25) + (-11.4535 \times 1) + (26.3020 \times 18) + (16.6372 \times 79) \\ &\quad + (-10.1897 \times 130) + (-2.1343 \times 64.8) + (-7.4969 \times 61) \\ &\quad + (8.3693 \times 2) + (22.0618 \times 4.1897) + (2.6037 \times 68) + 153.7365\end{aligned}$$

$$\begin{aligned}\hat{y}_2 &= (1.9413 \times 50) + (-11.4535 \times 2) + (26.3020 \times 28) + (16.6372 \times 103) \\ &\quad + (-10.1897 \times 229) + (-2.1343 \times 162.2) + (-7.4969 \times 60) \\ &\quad + (8.3693 \times 4.5) + (22.0618 \times 6.107) + (2.6037 \times 124) + 153.7365\end{aligned}$$

Q6 - Try the code with several learning rates that differ by orders of magnitude, and record the training and test set errors. Describe the theory of how changing the learning rate affects learning. What do you observe in the training error? How about the error on the test set?

The learning rate controls how quickly the model converges during gradient descent. In other words, in gradient descent, the learning rate controls the step size when updating the weights.

When the learning rate is too small, the cost decreases slowly, and the model may underfit if stopped early. When the learning rate is too large, the updates may overshoot the minimum and never converge. However, with the optimal learning rate, the cost decreases smoothly and efficiently reaches the minimum.

I recorded the train and set errors with several learning rates: 0.0001, 0.001, 0.01, 0.1, and 0.5. Additionally, I plotted the training cost (MSE) vs Iteration and the Predicted vs Actual values to see the effects of different learning rates visually (Figure 1).

- Learning rate = 0.0001: Train MSE = 27855.7246, Test MSE = 25102.2324. Too small, slow convergence, underfits and predicted points far from perfect fit line.
- Learning rate = 0.001: Train MSE = 19769.7011, Test MSE = 18346.2421. Slow convergence, still underfits.

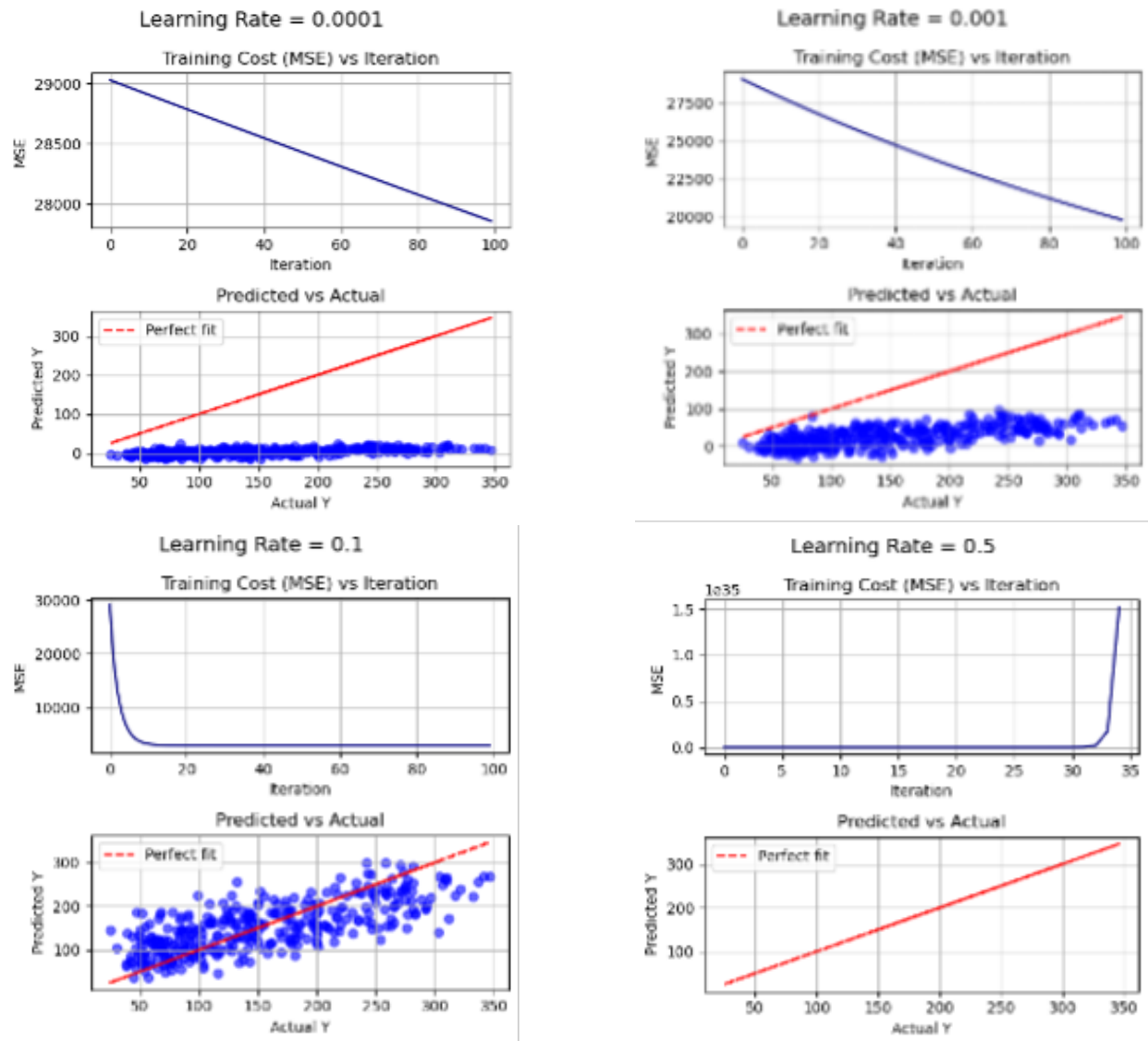


Figure 1: MSE vs Iteration + Predicted vs Actual for diff values of learning rate

- Learning rate = 0.1: Train MSE = 2890.0229, Test MSE = 2886.1491. Fast convergence (10 iterations), predicted points align with perfect fit line.
- Learning rate = 0.5: Train MSE = 15608.5117, Test MSE = nan. Too large, diverges and model fails.

Q8 - First of all, find the best value of alpha to use in order to optimize best. Next, experiment with different values of λ and see how this affects the shape of the hypothesis.

I changed the alpha value and observed the plot to find one where the cost starts high and decreases steadily.

- Learning rate = 0.001: Minimum cost: 3,4083. Cost decreases very slowly.
- Learning rate = 0.01: Minimum cost: 1.2585. Still decreases slowly, but better.
- Learning rate = 0.1: Minimum cost: 0.0551. Converges fast and has smooth convergence.
- Learning rate = 1: Minimum cost: 0.0164. Converges extremely fast with the lowest cost but might overshoot, so a better option is Lr=0.1.

After experimenting with different values of lambda, I did not observe a visible difference.